

Coding & Solution Implementation Report

VANGUARD CREDIT AUTOMATION | ACADEMIC PROJECT TRACK DELIVERABLE

1. Backend API Routing

Implemented in Python Flask (app.py). It registers routes for single predictions (/api/predict), batch compliance audits (/api/predict_batch), and algorithm statistics (/api/model_stats).

2. Training Pipeline

Implemented in train_models.py. It generates the synthetic credit applicant dataset, encodes categories, scales numerical values, trains models, evaluates metrics, and serializes artifacts.

3. Cloud Deployment Script

Implemented in watson_pipeline.py. It demonstrates deploying model artifacts to IBM Cloud Watson ML.